

FORT KNOX, KY, USA

WMO: 724240

Lat: 37.900N

Lon: 85.967W

Elev: 230

StdP: 98.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-10.4	-18.3	0.8	-11.9	-15.5	1.0	-9.2	10.2	8.2	9.1	6.2	2.7	280	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	33.6	24.3	32.3	24.1	31.1	23.6	26.1	31.2	25.3	30.3	24.6	29.3	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.3	28.7	23.9	19.3	27.9	23.1	18.3	26.9	82.0	31.2	78.6	30.3	75.5	29.3	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.8	-18.0	35.2	4.2	2.1	-21.0	36.7	-23.5	38.0	-25.9	39.2	-29.0	40.8		
(5)			-18.3	27.3	4.2	1.2	-21.3	28.1	-23.7	28.8	-26.0	29.5	-29.0	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	0.5	3.4	7.9	13.8	18.9	23.1	24.6	24.3	20.7	14.6	8.0	3.1
	DBStd	9.78	6.87	6.49	6.04	5.12	4.28	3.05	2.65	2.68	3.91	4.85	5.39	5.90
	HDD10.0	975	303	199	113	21	1	0	0	0	0	13	100	225
	HDD18.3	2440	552	417	328	150	48	4	1	1	22	136	311	472
	CDD10.0	2302	9	15	47	137	277	393	453	442	320	156	41	12
	CDD18.3	722	0	1	3	15	66	147	196	184	91	19	1	0
	CDH23.3	6444	0	1	23	132	545	1309	1788	1659	807	172	7	0
	CDH26.7	2332	0	0	2	18	139	471	714	657	293	37	1	0
Wind	WSAvg	3.0	3.7	3.6	3.6	3.7	2.9	2.5	2.3	2.2	2.3	2.7	3.1	3.4
Precipitation	PrecAvg	1227	92	85	116	121	139	106	109	84	75	91	87	112
	PrecMax	1813	160	183	375	443	286	235	192	182	253	228	205	276
	PrecMin	924	26	32	43	33	46	19	14	23	6	0	23	39
	PrecStd	197	35	39	65	74	56	50	43	44	58	59	42	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	21.7	26.2	28.6	31.0	33.4	36.5	35.7	34.6	30.1	24.0	20.0
		MCWB	15.6	15.3	17.9	18.6	22.2	24.7	24.9	24.8	21.6	21.4	17.1	16.4
	2%	DB	16.7	18.7	22.9	26.9	29.6	32.3	33.3	33.1	32.2	27.4	21.2	17.3
		MCWB	14.6	14.1	15.4	18.0	21.2	24.2	25.3	24.1	22.6	19.4	15.4	14.7
	5%	DB	14.1	16.4	20.0	24.8	28.4	31.0	32.0	31.8	30.1	25.1	18.8	14.7
		MCWB	12.0	12.7	13.7	16.8	20.7	23.5	24.9	23.8	21.8	18.0	14.6	12.4
	10%	DB	11.3	13.8	17.4	22.6	27.0	29.7	30.6	30.3	28.0	22.7	16.8	12.4
		MCWB	9.0	10.4	12.3	15.9	20.3	22.9	24.1	23.1	20.8	17.0	13.3	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	17.2	19.3	20.9	24.0	26.5	27.6	26.9	24.8	22.9	18.9	17.3
		MCDB	17.9	19.9	24.6	25.7	28.7	31.6	31.8	32.3	30.4	28.3	21.8	19.3
	2%	WB	14.9	15.3	16.9	19.7	22.8	25.4	26.3	25.7	23.9	21.2	17.2	15.2
		MCDB	16.5	17.7	20.6	24.3	27.7	30.3	31.3	30.9	29.2	24.6	19.3	16.9
	5%	WB	12.6	13.5	15.2	18.4	22.0	24.6	25.6	25.0	23.0	19.7	15.4	12.8
		MCDB	14.2	15.6	18.6	22.4	26.7	29.3	30.4	29.9	27.6	23.5	18.3	14.1
	10%	WB	9.2	10.6	13.2	17.0	21.0	23.7	24.8	24.1	22.2	18.2	13.5	10.4
		MCDB	11.1	13.1	16.7	21.1	25.0	28.3	29.4	28.6	26.3	21.6	16.4	12.0
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.0	10.9	12.0	10.9	10.7	10.3	11.0	11.8	11.9	10.3	8.6
		MCDBR	10.7	13.3	13.8	13.7	12.6	12.1	11.9	12.3	13.9	13.8	12.6	10.9
		MCWBR	8.8	10.1	8.4	7.3	5.8	5.3	4.8	4.8	5.6	6.9	8.4	8.9
	5% WB	MCDBR	9.8	12.0	11.8	11.2	10.6	10.6	10.7	10.6	11.1	11.2	10.8	10.0
		MCWBR	8.6	10.4	8.5	7.3	5.5	5.1	4.9	4.7	5.1	6.5	8.4	9.2
Clear-Sky Solar Irradiance	taub	0.310	0.323	0.354	0.403	0.441	0.455	0.481	0.484	0.419	0.359	0.329	0.306	
	taud	2.475	2.413	2.394	2.255	2.189	2.183	2.122	2.104	2.266	2.435	2.470	2.493	
	Ebn at Noon	875	907	907	877	845	830	805	792	828	853	843	854	
	Edh at Noon	81	98	110	134	146	147	155	154	124	95	81	75	
All-Sky Solar Radiation	RadAvg	1.95	2.64	3.72	4.94	5.51	6.24	5.92	5.52	4.57	3.42	2.35	1.68	
	RadStd	0.21	0.36	0.35	0.35	0.50	0.35	0.35	0.32	0.42	0.37	0.26	0.17	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+78

Nomenclature: See separate page